Nafisa Islam Rifa

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Nafisa Islam Rifa | NafisaIslamRifa | Portfolio | ` Google Scholar

London , United Kingdom

EDUCATION

- **University Of Greenwich** *September 2025 - October 2026 (Expected)*
  MSc in Data Science London, United Kingdom
  - Overall average: 75.8% (Distinction); Dissertation: 85%
- **United International University** *February 2022 - Aug 2025*
  Bachelor of Computer Science and Engineering Dhaka, Bangladesh
  - GPA: 3.95/4.00
- **Rajshahi College** *May 2019*
  Higher Secondary Certificate Rajshahi, Bangladesh
  - GPA: 5.00/5.00
- **Govt PN Girls' High School** *March 2017*
  Secondary School Certificate Rajshahi, Bangladesh
  - GPA: 5.00/5.00

PUBLICATIONS

C=CONFERENCE, J=JOURNAL, P=PATENT, S=IN SUBMISSION, T=THESIS

[C.1] (Accepted) *Iftekhar Mahmud, Puja Rani Das,* ***Nafisa Islam Rifa****,Irfan Hossain, Raiyan Rahman and Md. Farid Dewan*, Multimodal Emotion Recognition Using Visual and Thermal Image Fusion: A Deep Learning Approach, IEEE ICCIT 2024 [IEEE Conference].

- Worked on multimodal fusion of visual and thermal images
- Performed dataset preprocessing and model experimentation

[C.2] (Accepted) ***Nafisa Islam Rifa***, Visual Grounding and Explainability for Prompt-Driven Radiology Report Generation, 7th International Conference on Medical Imaging and Computer-Aided Diagnosis (MICAD 2026).

- Developed a parameter-efficient radiology vision-language model combining frozen Rad-DINO, a spatial adapter, and a 4-bit LoRA-adapted Qwen2.5-3B decoder.
- Introduced explicit coordinate-token supervision for jointly generating radiology reports and anatomically grounded bounding boxes.
- Developed an evaluation framework to audit visual grounding and explainability by comparing generated localizations with annotated regions and CAM- and prior-based baselines.

[J.1] (Under Review) ***Nafisa Islam Rifa*** *and Mohammad Nomani Kabir*, Video Moment Retrieval: A Survey of Methods, Benchmarks, and Open Challenges in the Multimodal LLM Era , Computer Vision and Image Understanding (Elsevier), (Q2, IF: 3.6).

- Surveyed 165 papers (2017–2026) and developed a unified taxonomy covering pre-MLLM, Transformer/VLP, and MLLM-era methods across five dimensions.
- Audited 34 MLLM-based methods, identifying gaps in bias-corrected evaluation, Video Corpus Moment Retrieval, and spatio-temporal grounding.
- Presented cross-era performance comparisons on major VMR benchmarks, along with consolidated results for VCMR and spatio-temporal grounding.

Projects

- **Build Large Language Model from Scratch** *Sep 2025*
  - Implemented end-to-end transformer components (tokenizer, embeddings, positional encodings, multi-head self-attention, feed-forward blocks and layer-norm) and assembled training with inference pipelines in PyTorch.
  - Trained and validated on a curated text corpus; implemented AdamW, LR scheduling, gradient accumulation and mixed-precision; evaluated with perplexity and qualitative sample generation.
  - Evaluated model performance on next-token prediction and small downstream tasks.
- **UKNest: Agentic RAG Assistant for UK Newcomers** *Sep 2026 – Oct 2026*
  - Built an **agentic RAG** assistant over 15 **GOV.UK** guides, with cited sources and update dates.
  - Exposed **Qdrant**, postcode and OpenStreetMap search through an **MCP** server with safety guardrails.
  - Achieved **1.00 Recall@5, MRR and tool-selection accuracy**; deployed with **Docker Compose**, GitHub Actions and Streamlit.
- **Agentic RAG London Tube Assistant with Live TfL Data** *Jul 2026 – Oct 2026*
  - Built an **LLM agent** combining live **TfL APIs** with RAG over official guidance using **fastembed** and **ChromaDB**.
  - Achieved **97% top-3 retrieval accuracy** and evaluated reranking, retaining vector search as the default.
  - Containerised with **Docker Compose**, tested with 95 unit tests and **GitHub Actions**, and deployed on Streamlit.
- **London Air Quality Forecaster: End-to-End MLOps** *Oct 2026*
  - Built an end-to-end **MLOps** pipeline using live **LAQN** and weather data, **DVC**, and **MLflow** on DagsHub.
  - Improved test MAE by **22.7%** over persistence using **LightGBM**; deployed a guarded **FastAPI** service with Docker and Render.
  - Implemented **CI/CD**, quality-gated retraining, and monitoring with **Prometheus**, **Grafana**, and GitHub Actions.

Skills

- **Programming Languages:** Python, R, SQL
- **Data Science & Machine Learning:** scikit-learn, PyTorch, TensorFlow, XGBoost, Regression, Classification, Clustering, Time-Series Analysis, Feature Engineering, PCA
- **Data Analysis & Wrangling:** Pandas, NumPy, Excel (Advanced)
- **Databases:** MySQL, MongoDB, Cassandra
- **Data Visualization:** Matplotlib, Seaborn, Plotly, Power BI, Tableau
- **Tools & Technologies:** Git, Roboflow, Figma

Services

- Reviewer, **MI4MedFM Workshop at MICCAI 2026**
- Reviewer, **WiML Workshop @ NeurIPS 2026**
- Reviewer, **Muslim in ML (MIML) Workshop @ NeurIPS 2026**

Language Proficiency

- **English:** Proficient, IELTS Academic – Overall Score: 7.5 (Listening: 8.0, Reading: 7.5, Writing: 6.5, Speaking: 7.5)
- **Bengali:** Native Language

References

Dr. Peter Soar
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